

MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 2 2025/2026

WEEK 5: CONTROLLING A 12V DC LINEAR ACTUATOR USING L298P MOTOR

DRIVER AND RELAY

SECTION 1

GROUP 9

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1.0 INTRODUCTION

The first task for this week's report focuses on the fundamental theory of direction control by requiring the design of a manual H-bridge circuit using four discrete switches. This task serves to illustrate the internal polarity-switching logic typically automated by the L298P motor driver, emphasizing how specific current paths are established to either extend or retract the actuator without the use of an integrated silicon driver.

The second task shifts the focus toward closed-loop automation using a 2-way channel relay module. In this task, a sensor is integrated into the system to monitor the actuator's position, providing real-time feedback to the Arduino UNO. The objective is to program the microcontroller to execute a safety interrupt that stops the actuator and activates an LED indicator once a specific detection point is reached, demonstrating a practical application of mechatronic system integration and safety signaling.

2.0 MATERIALS AND EQUIPMENT

2.1 Task 1

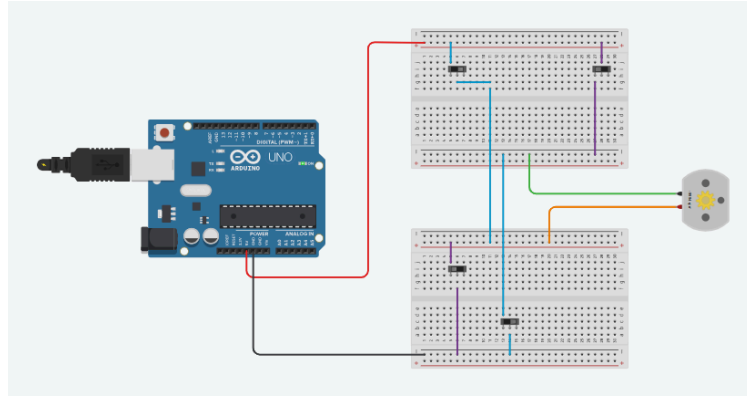
- Microcontroller: Arduino Uno and USB cable
- Input component: Push buttons
- Output component: DC motor
- Prototyping: Breadboard and Jumper wires

2.2 Task 2

- Microcontroller: Arduino Uno and USB cable
- Input component: DC power supply and Ultrasonic Sensor
- Output component: LED and 12V linear actuator
- Prototyping: 2-way channel relay and Jumper wires
- Software: Arduino IDE

3.0 EXPERIMENTAL SETUP

3.1 Task 1

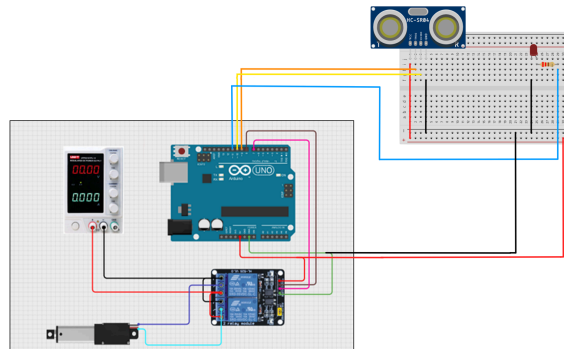


Picture 1: Connection Components for Task 1

1. Connection of the Arduino Uno to breadboards:
 - Connect the 5V pin to the positive terminal on breadboard 1.
 - Connect the GND pin to the negative terminal on breadboard 2.
2. Connection of blue wire:
 - Connect from 5V terminal to one of the switch 1 legs.
 - Connect the wire from the middle leg of the switch 1 to the negative polarity of the DC motor.
 - Connect the wire from the positive polarity of the DC motor to one of the switch 4 legs.
 - Connect the wire from the middle leg of the switch 4 to the GND terminal on the breadboard.
3. Connection of purple wire:
 - Connect the wire from the 5V terminal on the breadboard to one of the switch 2 legs.
 - Connect the wire from the middle leg of the switch 2 to the positive polarity of the DC motor.

- Connect the wire from the negative polarity of the DC motor to one of the switch 3 legs.
- Connect wire from the middle leg of the switch 3 to the GND terminal on the breadboard.

3.2 Task 2



Picture 2: Connection Components for Task 2

1. Connection of Arduino Uno and breadboard:
 - Connect the 5V pin to the positive terminal on the breadboard.
 - Connect the GND pin to the negative terminal on the breadboard.
2. Connection of Ultrasonic Sensor to Arduino Uno:
 - Connect the VCC pin at the ultrasonic sensor to the 5V terminal at the breadboard.
 - Connect the Trig pin at the ultrasonic sensor to the D9 pin on the arduino.
 - Connect the Echo pin at the ultrasonic sensor to the D10 pin on the arduino.
 - Connect the GND pin at the ultrasonic sensor to the GND terminal at the breadboard.
3. Connection of LED to Arduino Uno:

- Connect the cathode LED leg to the GND terminal at the breadboard.
 - Connect the anode LED leg to the 220Ω resistor.
 - Connect the 220Ω resistor to the D11 on the arduino.
4. Connection of 12V DC Linear Actuator to 2 Way Channel Relay:
- Connect the positive wire of the actuator to COM1 at relay 1.
 - Connect the negative wire of the actuator to COM2 at relay 2.
5. Connection of 12 DC Power Supply to 2 Way Channel Relay:
- Connect the positive wire to the NC1 and NC2 at the relay 1 and 2.
 - Connect the negative wire to the NO1 and NO2 at the relay 1 and 2.
6. Connection of 2 Way Channel Relay to Arduino Uno:
- Connect the VCC pin on the 2 channel relay to the 5V terminal at the breadboard.
 - Connect the GND pin on the 2 channel relay to the GND terminal at the breadboard.
 - Connect the IN2 pin on the 2 channel relay to the D8 pin on the arduino.
 - Connect the IN1 pin on the 2 channel relay to the D7 pin on the arduino.

4.0 METHODOLOGY

4.1 Task 1

4.1.1 Circuit Setup

- Interface the Arduino UNO with the L298P Motor Driver Shield, ensuring all headers are correctly aligned to prevent pin displacement.
- Connect the 12VDC Linear Actuator terminals to the Motor A (MA) output screw terminals and ensure the mechanical connection is secure.
- Remove the OPT jumper from the L298P shield to isolate high-current inductive loads from the logic circuitry and prevent damage to the USB port.
- Connect a variable DC power supply to the VMS and GND terminals, setting the output to a constant 12V DC with a 3.0A current limit.

4.1.2 Program Operation

- Turn on the Variable DC Power Supply and observe the actuator automatically extending and retracting based on the programmed 3000ms delays.
- Verify that the actuator reverses polarity correctly; if necessary, reverse the actuator wires to change the default orientation.

4.2 Task 2

4.2.1 Circuit Setup

- All the required components for this experiment are prepared before assembling the circuit.
- The 12V DC Power Supply positive wire is connected to NC1 and NC2 while the negative wire is connected to NO1 and NO2 on the 2 Way Channel Relay like in Picture 1.

- The 12V DC Linear Actuator positive wire is connected to COM1 and the negative wire is connected to COM2 on the 2 Way Channel Relay like in Picture 1.
- The 2 Way Channel Relay IN1 and IN2 pins are connected to the D7 and D8 pins on Arduino Uno. The VCC and GND pins on the 2 Way Channel Relay are connected to the 5V and GND terminal on the breadboard.
- The Ultrasonic Sensor Trig and Echo pins are connected to D9 and D10 pins on the Arduino Uno. The VCC and GND pins are connected to the 5V and GND terminal on the breadboard
- The LED anode leg is connected to 220 Ω resistor and then to D11 pin on Arduino Uno. The cathode LED leg is connected to the GND terminal on the breadboard.
- The Arduino Uno 5V pin is connected to the positive terminal and the GND pin is connected to the negative terminal on the breadboard.
- After the connection between components has been done, the wiring around the component was checked carefully to ensure the project is working properly and to avoid short circuits.

4.2.2 Program Development and Uploading

- Assign pins 7 and 8 to control the relay channels and define the input pin for the integrated sensor.
- Write subroutines for extendActuator() and retractActuator() by toggling relay states between HIGH and LOW.
- Implement an if statement to monitor sensor data so that detecting the actuator at a certain point triggers a stop command and activates the LED.

4.2.3 Program Operation

- Power the system and initiate the loop, allowing the actuator to move toward the detection point.
- Observe the system behavior when the sensor is triggered; the actuator should cease movement immediately as the LED light activates.

4.3.4 Coding

```
const int relay1 = 7;
const int relay2 = 8;
const int trigPin = 9;
const int echoPin = 10;
const int ledPin = 11;

const int stopDistance = 10;

void setup() {
    pinMode(relay1, OUTPUT);
    pinMode(relay2, OUTPUT);
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    pinMode(ledPin, OUTPUT);
    stopActuator();
}

void loop() {
    int distance = getDistance();

    if (distance > 0 && distance <= stopDistance) {

        stopActuator();
        digitalWrite(ledPin, HIGH);
    } else {

        digitalWrite(ledPin, LOW);

        extendActuator();
    }
}
```

```
        delay(100);
    }

    int getDistance() {
        digitalWrite(trigPin, LOW);
        delayMicroseconds(2);
        digitalWrite(trigPin, HIGH);
        delayMicroseconds(10);
        digitalWrite(trigPin, LOW);

        long duration = pulseIn(echoPin, HIGH, 30000);
        return duration * 0.034 / 2;
    }

    void extendActuator() {
        digitalWrite(relay1, HIGH);
        digitalWrite(relay2, LOW);
    }

    void stopActuator() {
        digitalWrite(relay1, LOW);
        digitalWrite(relay2, LOW);
    }
}
```

5.0 DATA COLLECTION

5.1 Task 1

Push Button Condition	Motor Rotation
Top left & bottom right	Clockwise
Bottom left & top right	Anti-clockwise
Otherwise	Does not move

5.2 Task 2

Sensor State	Actuator Action	LED Status
Not detected	Extending	OFF
Not detected	Retracting	OFF
Detected	Stopped	ON

6.0 DATA ANALYSIS

6.1 Task 1

The analysis of the manual H-bridge circuit demonstrates the fundamental relationship between switch configuration and the directional movement of the 12V DC linear actuator. By manually toggling the four-switch network, it was observed that closing specific diagonal pairs successfully reversed the polarity across the motor terminals. This mechanical simulation validates the inverted logic utilized by the L298P motor driver shield, confirming that bidirectional control is dependent on the path of current flow rather than the absolute voltage level.

6.2 Task 2

The data collected demonstrates successful closed-loop feedback integration within the mechatronic system. By interfacing a sensor with the Arduino UNO, the system transitioned from simple timed delays to event-driven logic, where the physical state of the actuator dictated the relay's operation. Analysis of the results shows that upon detection at the certain point specified in the task, the microcontroller immediately toggled the relay pins to a neutral or opposing state to halt the 12V DC linear actuator. Simultaneously, the activation of the LED provided immediate visual confirmation of the system's interrupt state, validating the reliability of the integrated sensor as a safety or positioning limit.

7.0 RESULTS

7.1 Task 1

The results in task 1 is that we can control the actuator without using the motor driven or arduino Uno. Using only wire and push button, we can make a h-bridge and make the actuator move forward and reverse. When the button 1 and 4 is pushed together, the actuator will move forward as shown in the diagram 7.1.1. On the other hand, when the button 2 and 3 is pushed, the actuator will move in reverse. If only 1 button is pushed, the actuator will not respond. Other than that, if button 1 and 2 is pushed, the actuator also will not move.

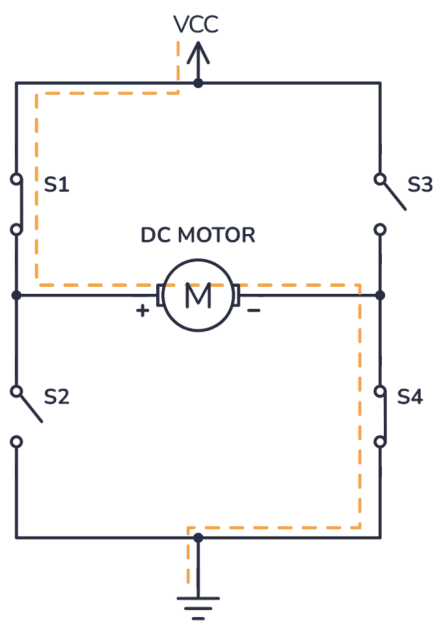


Diagram 7.1.1 show the movement of the current when button 1 and 2 is being pushed

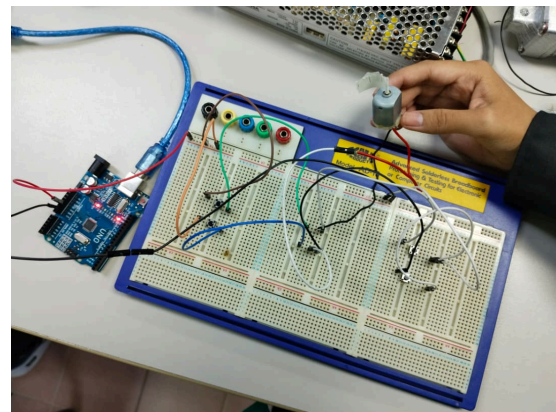


Diagram 7.1.2 show the H-Bridge circuit installation

7.2 Task 2

The results for experiment 2 is that the ultrasonic sensor successfully detected the movement of the actuator and measured the distance of the actuator in real time. After that, the system response in this experiment worked very well. When the actuator moves in front of the sensor. The actuator suddenly stops moving and the LED will turn on. This happens because the relay switched to the LOW/LOW state and made the actuator stop moving. On the other hand, when no object is detected, the actuator will keep moving and the LED stays off.

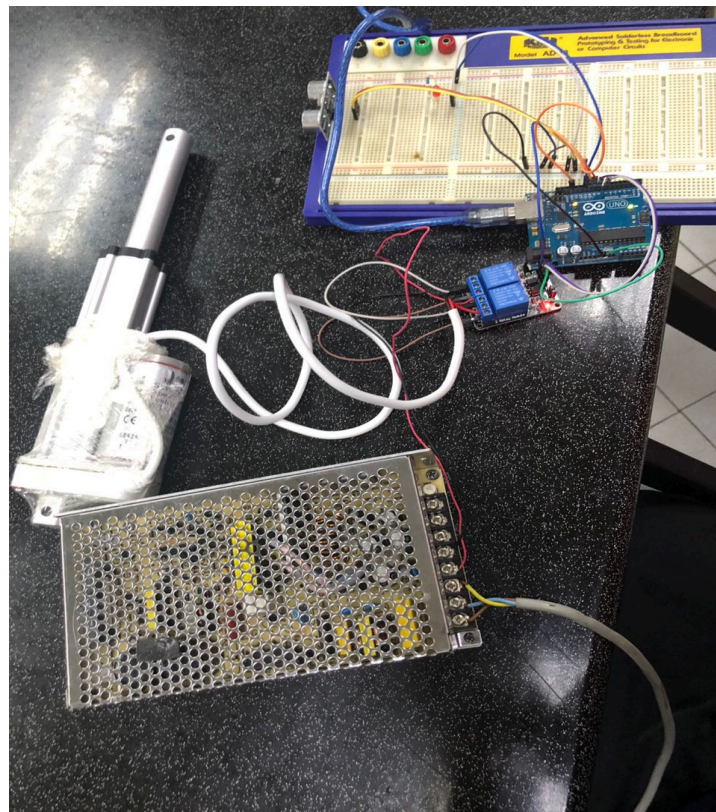


Diagram 7.2.1 shows the circuit installation of experiment 2

8.0 DISCUSSION

8.1 Task 1

Pulse Width Modulation (PWM) is a digital technique used to regulate the velocity of the 12V DC linear actuator by rapidly switching the motor's power supply ON and OFF. Rather than providing a steady, lower voltage, the Arduino UNO sends high-frequency signals via pins D10 or D11, where the duty cycle, the ratio of ON time to the total cycle time, determines the average voltage delivered to the actuator. For instance, a 50% duty cycle results in a medium speed, while a 100% duty cycle provides full power. By adjusting this duty cycle through the `analogWrite()` function (ranging from 0 to 255), you can achieve precise speed control, allowing the actuator to extend or retract at variable rates while maintaining the high-current efficiency required by the L298P driver.

The H bridge function is to make it move forward, reverse or brake. H-bridge function when we push button 1 and 4 simultaneously, then the actuator will move forward. This is because the current moves from the left side to the right side of the circuit. This will make the current flow from left to the right of the actuator. If the button 2 and 3 is pushed, the current will flow from the right side to the left side of the circuit. This makes the current enter the actuator from right to the left and make the actuator move in reverse. When no button is pushed then the current will not flow and make the actuator stop. If only 1 button is being pushed or 2 buttons from the same side are being pushed then the actuator will not move because the current flow is not complete.

8.2 Task 2

In this experiment, we use the `getDistance()` function in coding to calculate how far an object is. This function works by measuring the time it takes for a sound wave to bounce back. After that, in the coding, we use the void function in the loop function. So if the sensor detects the movement, it overrides the movement command and the actuator will stop moving. Lastly, the LED feedback is important and most standard features in mechatronics projects because of its function as visual indicators and can alert people.

9.0 CONCLUSION AND RECOMMENDATION

9.1 Task 1

To conclude for task 1 is that H bridge is a cheap and easy concept for mechatronics students to make their project. This is because the actuator can be controlled forward, reverse and can be stopped just using wires and push buttons. The users did not need to change the wire position to make the actuator move in a different direction. We just need to push different buttons and we can control the actuator. My recommendation for this experiment is to make sure the wiring is correct and tidy. Making H-bridge use a lot of wire and sometimes connections can be wrong. Other than that, we can make a time delay using an arduino so the actuator can be used longer and will not break easily.

9.2 Task 2

For conclusion, this task successfully demonstrates how to integrate a sensor with a relay-controlled actuator system. This system manages to automate the stopping process by using an ultrasonic sensor to detect a specific point. This system is widely used in the industry as the safety features for their big operation machine. The recommendation for this experiment is to use PWM for actuator movement, so we can set it to move slower as it

comes closer to the ultrasonic sensor. Other than that, we can add a buffer to make the stopping smoother so the relays do not click rapidly when the object is detected.

11.0 REFERENCES

Tutorial for L298 2Amp Motor Driver Shield for Arduino

<https://www.instructables.com/Tutorial-for-L298-2Amp-Motor-Driver-Shield-for-Ard/>

How to Use L298P drive shield: Examples, Pinouts, and Specs

<https://docs.cirkitdesigner.com/component/a25f6e70-294e-42fd-b77c-9e9349b76b9a/l298p-drive-shield>

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STUDENTS' DECLARATION

We hereby declare that the work presented in this technical report is entirely our own, except where explicitly acknowledged. We certify that in developing and completing this mechatronics project, we have strictly adhered to the standards of academic integrity and have not engaged in any form of plagiarism or unethical conduct. All sources of information, reference materials, and technical support utilized in this work have been appropriately cited and acknowledged.

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been untaken or done by unspecified sources or persons. We hereby certify that this report has not been done by only one individual, and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have read and understand the content of the total report, and no further improvement on the reports is needed from any of the individual contributors to the report. We therefore agreed

unanimously that this report shall be submitted for marking, and this final printed report has been verified by us.

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